SE LAB 7
PASSWORD POLICY

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```
C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>python -m pip install hypothesis py
test
Collecting hypothesis
  Downloading hypothesis-6.170.15-cp315-abi3.abi3t-win_amd64.whl.metadata (5.6 kB)
Requirement already satisfied: pytest in C:\Python315\Lib\site-packages (9.2.0)
Collecting sortedcontainers<3.0.0,>=2.1.0 (from hypothesis)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: colorama>=0.4 in C:\Python315\Lib\site-packages (from pytest) (0.4.6)
Requirement already satisfied: iniconfig>=2 in C:\Python315\Lib\site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in C:\Python315\Lib\site-packages (from pytest) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in C:\Python315\Lib\site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in C:\Python315\Lib\site-packages (from pytest) (2.21.0)
Downloading hypothesis-6.170.15-cp315-abi3.abi3t-win_amd64.whl (691 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 691.0/691.0 kB 927.8 kB/s  0:00:00
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Installing collected packages: sortedcontainers, hypothesis
Successfully installed hypothesis-6.170.15 sortedcontainers-2.4.0

[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>python -m pytest --version
pytest 9.2.0

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>python -m pip show hypothesis
Name: hypothesis
Version: 6.170.15
Summary: The property-based testing library for Python
Home-page: https://hypothesis.works
Author:
Author-email: "David R. MacIver and Zac Hatfield-Dodds" <david@drmaciver.com>
License-Expression: MPL-2.0
Location: C:\Python315\Lib\site-packages
Requires: sortedcontainers
Required-by:
```

```
C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>python password_policy.py
==== Running first_character_type ====
Enter a password: Abc123!
First Character Type: letter

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): alice:Secret123
User and Password: ('alice', 'Secret123')

==== Running digit_ratio ====
Enter a password to measure: abc123
Digit Ratio: 0.5
```

```
def first_character_type(pw):
    if not pw:
        return "empty"

    ch = pw[0]

    if ch.isdigit():
        return "digit"
    if ch.isalpha():
        return "letter"
    return "symbol"


def split_credentials(text):
    if not isinstance(text, str):
        return "", ""

    if ":" not in text:
        return text.strip(), ""

    user, password = text.split(":", 1)
    return user.strip(), password.strip()


def digit_ratio(pw):
    if not isinstance(pw, str) or len(pw) == 0:
        return 0.0

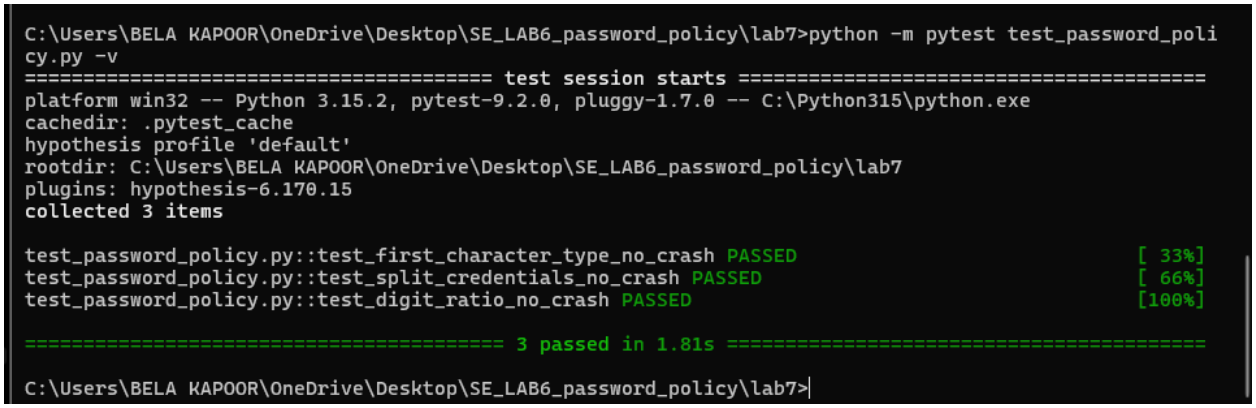
    return sum(c.isdigit() for c in pw) / len(pw)

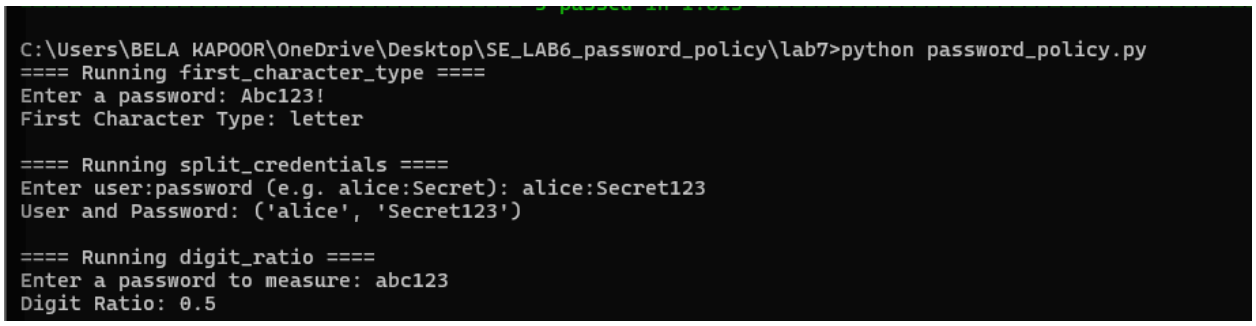

def main():
    print("==== Running first_character_type ====")
    user_input = input("Enter a password: ")
    result = first_character_type(user_input)
    print("First Character Type:", result)

    print("\n==== Running split_credentials ====")
    user_input = input("Enter user:password (e.g. alice:Secret): ")
    result = split_credentials(user_input)
    print("User and Password:", result)

    print("\n==== Running digit_ratio ====")
    user_input = input("Enter a password to measure: ")
    result = digit_ratio(user_input)
    print("Digit Ratio:", result)


if __name__ == "__main__":
    main()
```





Conclusion

In this lab, fuzz testing was performed on a Python password policy module using the Hypothesis library. Automated test cases were used to identify errors caused by unexpected inputs such as empty strings, None values, and incorrectly formatted credentials.

The identified errors were fixed by implementing proper input validation and handling edge cases. The test cases were executed again using Pytest to verify the corrected implementation.

This experiment demonstrated how fuzz testing helps identify hidden bugs and improve the reliability and robustness of software.

```
from hypothesis import given, strategies as st, example
from password_policy import first_character_type, split_credentials, digit_ratio


@given(st.text() | st.none())
@example("")
@example(None)
def test_first_character_type_no_crash(s):
    first_character_type(s)


@given(st.text() | st.none())
@example("")
@example(None)
@example("alice")
@example("alice:one:two")
def test_split_credentials_no_crash(s):
    split_credentials(s)


@given(st.text() | st.none())
@example("")
@example(None)
def test_digit_ratio_no_crash(s):
    digit_ratio(s)
```

```
C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>python -m pytest test_password_policy.py -v
========================================= test session starts =========================================
platform win32 -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Python315\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7
plugins: hypothesis-6.170.15
collected 3 items

test_password_policy.py::test_first_character_type_no_crash FAILED                           [ 33%]
test_password_policy.py::test_split_credentials_no_crash FAILED                              [ 66%]
test_password_policy.py::test_digit_ratio_no_crash FAILED                                    [100%]

============================================== FAILURES ===============================================
_____________________________ test_first_character_type_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 7, in test_first_character_type_no_crash
  |     @example("")
  |
  |   File "C:\Python315\Lib\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 10, in test_first_character_type_no_crash
    |     first_character_type(s)
    |     ~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 2, in first_character_type
    |     ch = pw[0]
    |          ~~^^^
    | IndexError: string index out of range
    | Failing explicit example: test_first_character_type_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 10, in test_first_character_type_no_crash
    |     first_character_type(s)
    |     ~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 2, in first_character_type
    |     ch = pw[0]
    |          ~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    | Failing explicit example: test_first_character_type_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_split_credentials_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 14, in test_split_credentials_no_crash
  |     @example("")
  |
  |   File "C:\Python315\Lib\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
  |     raise the_error_hypothesis_found
```

```
|
|   File "C:\Python315\Lib\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 19, in test_split_credentials_no_crash
  |     split_credentials(s)
  |     ~~~~~~~~~~~~~~~~~~^^^
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 10, in split_credentials
  |     user, password = text.split(':')
  |     ^^^^^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  | (note: 2 other explicit examples also failed with this error; use Verbosity.verbose to view)
  | Failing explicit example: test_split_credentials_no_crash(
  |     s='',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 19, in test_split_credentials_no_crash
  |     split_credentials(s)
  |     ~~~~~~~~~~~~~~~~~~^^^
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 10, in split_credentials
  |     user, password = text.split(':')
  |                      ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  | Failing explicit example: test_split_credentials_no_crash(
  |     s=None,
  | )
  +------------------------------------
_____________________________ test_digit_ratio_no_crash ______________________________
+ Exception Group Traceback (most recent call last):
|   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 23, in test_digit_ratio_no_crash
|     @example("")
|
|   File "C:\Python315\Lib\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 26, in test_digit_ratio_no_crash
  |     digit_ratio(s)
  |     ~~~~~~~~~~~^^^
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 14, in digit_ratio
  |     return sum(c.isdigit() for c in pw) / len(pw)
  |            ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~
  | ZeroDivisionError: division by zero
  | Failing explicit example: test_digit_ratio_no_crash(
  |     s='',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 26, in test_digit_ratio_no_crash
  |     digit_ratio(s)
  |     ~~~~~~~~~~~^^^
  |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 14, in digit_ratio
```

```
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\test_password_policy.py", line 26, in test_digit_ratio_no_crash
    |     digit_ratio(s)
    |     ~~~~~~~~~~~^^^
    |   File "C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7\password_policy.py", line 14, in digit_ratio
    |     return sum(c.isdigit() for c in pw) / len(pw)
    |                ^^^^^^^^^^^^^^^^^^^^^^^^
    | TypeError: 'NoneType' object is not iterable
    | Failing explicit example: test_digit_ratio_no_crash(
    |     s=None,
    | )
    +------------------------------------
=================================== short test summary info ====================================
FAILED test_password_policy.py::test_first_character_type_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
FAILED test_password_policy.py::test_split_credentials_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
FAILED test_password_policy.py::test_digit_ratio_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions)
======================================== 3 failed in 4.53s =========================================

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab7>
```